Number Of Ways To Traverse Graph

AlgoExpert · Medium · Dynamic Programming

The problem

- Paths from top-left to bottom-right of a width x height grid moving only right/down.
- Equals C(w+h-2, w-1), computed incrementally to avoid overflow.

Intuition

- Paths to a cell come from above or from the left; this is also a binomial coefficient.

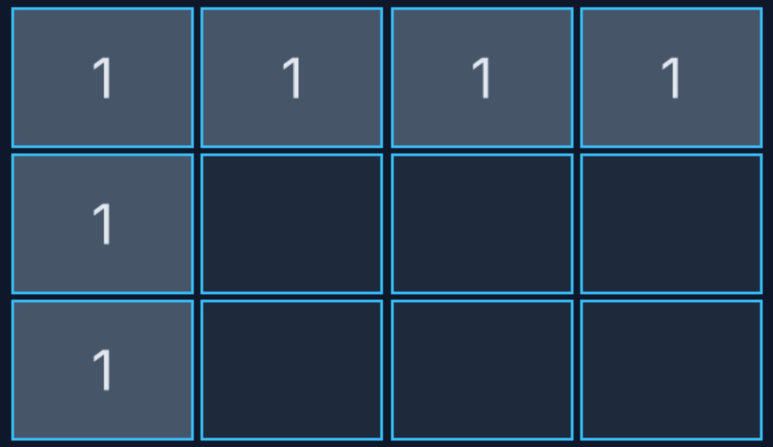
Brute force vs optimal

- Recursion: ways(r,c) = ways(r−1,c) + ways(r,c−1) → exponential
- DP grid → O(w·h)
- Combinatorics: C(w+h−2, h−1) → O(min(w,h))

Approach

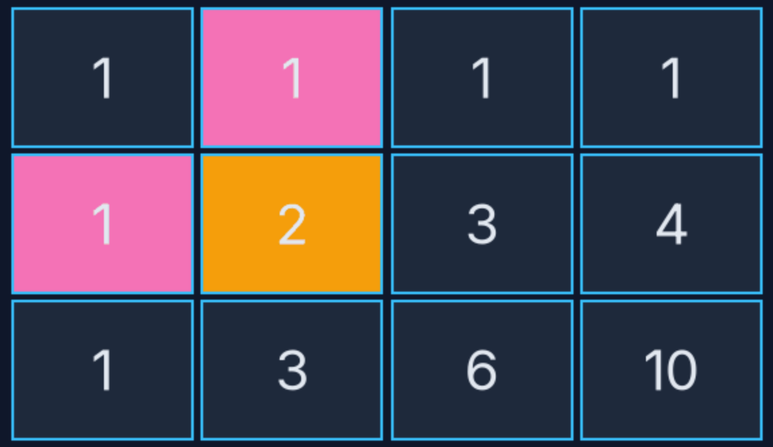
- 1. First row and column have one way
- 2. Each cell sums its top and left neighbours
- 3. Return the bottom-right cell

Worked example



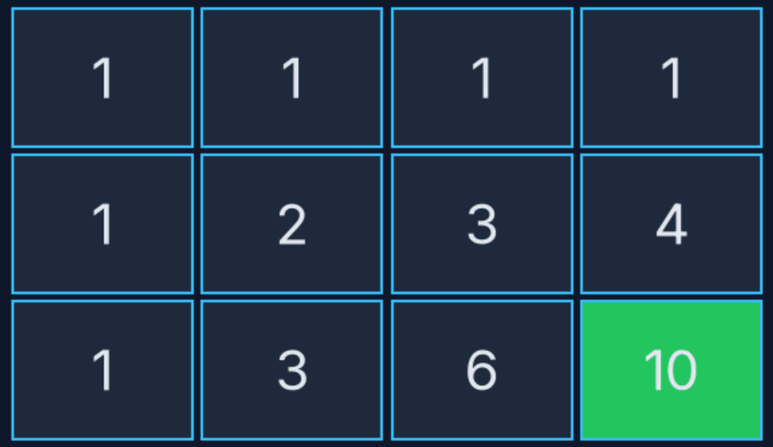
Edges: exactly one way (straight line)

Worked example



Each cell = above + left

Worked example



Bottom-right = 10

Worked example

```
moves: 3 right + 2 down = 5

choose which 2 are 'down': C(5, 2) = 10
```

Shortcut: a binomial coefficient

Java solution

```java
public class NumberOfWaysToTraverseGraph {
    public static long numberOfWaysToTraverseGraph(int width, int height) {
        int n = width + height - 2, k = Math.min(width, height) - 1;
        long r = 1;
        for (int i = 1; i <= k; i++) r = r * (n - k + i) / i;
        return r;
    }
}
```

Complexity

- Time: O(min(w, h))
- Space: O(1)

Edge cases & pitfalls

- 1×n grids → 1
- Compute C(n,k) incrementally to avoid overflow
- Width and height are counts, not indices

Interview tips & follow-ups

- Present DP first, then the formula
- Follow-up: obstacles → DP is required

Recap

- Paths to a cell come from above or from the left; this is also a binomial coefficient.
- Complexity: O(min(w, h))